

Homework 5

AMATH 582, Winter 2026

Due on Mar 16, 2026 at midnight.

DIRECTIONS, REMINDERS AND POLICIES

Read these instructions carefully:

- You are required to upload a PDF report to Gradescope along with your code in the form of a Jupyter notebook or a zip file of your main scripts and functions. The report and the code are to be submitted under separate assignments. **DO NOT INCLUDE YOUR CODE IN THE REPORT.**
- The report should be a maximum of 10 pages in accordance with the template on Canvas including references. Minimum font size 11pts and margins of at least 1inch on A4 or standard letter size paper. The report should be formatted as follows:
 - Title/author/abstract: Title, author/address lines, and short (100 words or less) abstract. This is not meant to be a separate title page.
 - Sec. 1. Introduction and Overview
 - Sec. 2. Theoretical Background
 - Sec. 3. Algorithm Implementation and Development
 - Sec. 4. Computational Results
 - Sec. 5. Summary and Conclusions
 - Acknowledgments. Mention other students if you were part of a study group.
 - References
- **Recall our late policy: If you plan to use your tokens please send an email to Saba (heravi@uw.edu). If you do not use a token you will lose 1/3 of your grade for each day the submission is late. WE WILL NOT AUTOMATICALLY APPLY TOKENS. YOU MUST INFORM US IF YOU PLAN TO USE THEM.**

PROBLEM DESCRIPTION: CLASSIFYING POLITICIANS

Your goal is to test the performance of spectral clustering and a simple semi-supervised regression algorithm on the 1984 house voting records data set. Download the data set called `house-votes-84.data`, and the description `house-votes-84.names`. The data set consists of voting records of 435 members of the House on 16 bills. There are 267 members of the democratic party and 168 members of the republican party. The voting record of each house member on the 16 bills will be our input $\mathbf{x}$ while the corresponding output/class y is that members party affiliation (republican or democrat embedded as ± 1).

TASKS

Below is a list of tasks to complete in this homework and discuss in your report.

1. Your first task is to import and preprocess the data set. Construct your output vector $\mathbf{y}$ by assigning labels $\{-1, +1\}$ to members of different parties. Then construct the input vectors $\mathbf{x}_j$ corresponding to the voting records of each member by replacing 'y' votes with +1, 'n' votes with -1 and '?' with 0. You do not need to center and normalize the data set in this case. This leads to a vector $\mathbf{y} \in \mathbb{R}^{435}$ and input matrix $X \in \mathbb{R}^{435 \times 16}$. Note that we are using the `sklearn` convention for our X matrix.
2. (**Spectral Clustering**): In this step you will mainly work with the matrix X and use $\mathbf{y}$ for validation of your clustering algorithm.

- (a) Construct the unnormalized graph Laplacian matrix on X using the weight function

$$\eta(t) = \exp\left(-\frac{t^2}{2\sigma^2}\right),$$

with variance parameter σ and compute its second eigenvector (i.e., the Fiedler vector) which we denote as $\mathbf{q}_1$.

- (b) Take $\text{sign}(\mathbf{q}_1)$ as your classifier and compute its classification accuracy after comparison with $\mathbf{y}$:

$$\text{clustering accuracy} = 1 - \frac{1}{435} \times \text{number of misclassified members.}$$

Note 1: since this is an unsupervised learning task, your classifier $\text{sign}(\mathbf{q}_1)$ might disagree with $\mathbf{y}$ by a negative sign, that is, the +1 class may be assigned to -1 and vice versa. Don't forget to adjust for this.

Note 2: "misclassified" here means a member who should have belonged to the opposite party/cluster.

- (c) Change the parameter σ in an appropriate range (e.g. $(0, 4]$) and plot accuracy as a function of σ . Let σ^* denote the optimal variance parameter achieving maximum clustering accuracy. Report this value and discuss your findings.
3. (**Semi-supervised Learning**): Now consider the unnormalized Laplacian from Step 2 with the optimal parameter σ^* that you picked in step 2c.

- (a) Given an integer $M \geq 1$ consider the Laplacian embedding

$$F(\mathbf{x}_j) = \left((\mathbf{q}_0)_j, (\mathbf{q}_1)_j, \dots, (\mathbf{q}_{M-1})_j \right) \in \mathbb{R}^M,$$

where $\mathbf{q}_j$ denote the eigenvectors of the Laplacian matrix. Write $F(X) \in \mathbb{R}^{435 \times M}$ for the Laplacian embedding of X , i.e., the matrix whose j -th row is $F(\mathbf{x}_j)$.

- (b) Given an integer $J \geq 1$ consider the submatrix $A \in \mathbb{R}^{J \times M}$ and vector $\mathbf{b} \in \mathbb{R}^J$ consisting of the first J rows of $F(X)$ and $\mathbf{y}$,

$$\begin{aligned} A_{ij} &= F(X)_{ij}, \quad i = 0, \dots, J-1, \quad j = 0, \dots, M-1, \\ \mathbf{b}_i &= \mathbf{y}_i, \quad i = 0, \dots, J-1. \end{aligned}$$

Warning: It is crucial here to use the original ordering of the data set. You should not be shuffling or re-ordering the rows of X .

- (c) Use linear regression (least squares) to find

$$\hat{\boldsymbol{\beta}} = \operatorname{argmin}_{\boldsymbol{\beta} \in \mathbb{R}^M} \|A\boldsymbol{\beta} - \mathbf{b}\|_2^2,$$

then take $\hat{\mathbf{y}} = \operatorname{sign}(F(X)\hat{\boldsymbol{\beta}})$ as the predictor of classes of all points in X .

Note: You do not need to add the column of 1's in A since $\mathbf{q}_0$ is already a constant vector so that β_0 accounts for the intercept.

- (d) Provide a table summarizing the accuracy of $\hat{\mathbf{y}}$ as your classifier for $M = 2, 3, 4, 5, 6$ and $J = 5, 10, 20, 40$.

$$\text{SSL accuracy} = 1 - \frac{1}{435} \times \text{number of misclassified members.}$$

Discuss your findings.